Astronomical Anomaly Detection with Spiking Neural Networks

Hannah Garcia
Washington State University
hannah.k.garcia@wsu.edu
011854857

Akalya Sridharan
Washington State University
akalya.sridharan@wsu.edu
011795984

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Abstract

In modern astronomical surveys, the generated time series data is extensive, and infrequent occurrences are indicative of important phenomena. Identifying anomalies is difficult because they are scarce, complex, and often lack sufficient annotated data for a comprehensive understanding. This project proposes a semi-supervised anomaly detection approach based on spiking neural networks (SNNs) for characterizing normal behavior in temporal signals and identifying anomalies indicative of anomalous events. Astronomical continuous light curves are represented as spike trains to enhance changes and enable the SNN to learn normal behavior. Synthetic time series signal data with anomalies are used for experiments to quantitatively measure performance. To prove the applicability of the proposed approach, it's tested on the available Kepler light curve data. Performance is evaluated based on detection rate, false alarm rate, and anomaly scoring consistency. The project's objective is to evaluate the effectiveness of SNNs for temporal anomaly detection in large-scale astronomical datasets.

1 Introduction

Time-domain astronomy is defined as the study of how astronomical objects change over time, encompassing any naturally occurring physical entities within the universe. From star clusters to galaxies, these systems consist of multiple astronomical bodies or substructures, such as asteroids or planets. The continuously evolving technology involved in astronomy has made the study of time-domain astronomy more feasible and important. Changes that were once too minuscule to observe can now be captured with high temporal resolution. This has permitted researchers to make a step forward in studying rare anomalies, as well as processes related to stellar evolution and planetary dynamics, from events that occur within a span of milliseconds to years [1]. Due to the extremely large temporal datasets in astronomy, this topic presents an opportunity to apply machine learning methods to efficiently identify anomalous patterns.

Anomaly detection plays a key role in astronomical surveys, which often produce datasets of unparalleled scale and greatly increase the probability of scientific discovery. Due to the sheer number and variance of sources, it is impractical to manually inspect or label each piece of data. As a result, traditional supervised learning approaches are infeasible. Methods capable of learning useful representations of astronomical anomalies have thus grown to become increasingly important [2].

Spiking neural networks (SNNs), in particular, provide a basis that mimics biological neural networks. SNNs have been built on artificial neural networks (ANNs) and deep neural networks (DNNs), representing information through discrete spike timings instead of continuous activations [3]. This structure creates an efficient modeling of sparse, time-varying signals that are found in astronomical observations such as light curves and RFI measurements [4]. SNNs have the capacity for efficient, real-time detection of anomalous patterns, which are overlooked by traditional machine learning methods.

2 Related Work

Recent advances in machine learning have shown tremendous progress in the study of astronomical data. For example, Muthukrishna et al. developed RAPID, a real-time automated photometric time-series classification tool that identifies transient phenomena. RAPID employs deep recurrent neural networks with gated recurrent units for classifying astronomical time-series data. RAPID can be applied to ongoing surveys such as the Zwicky Transient Facility (ZTF) and the Large Synoptic Survey Telescope (LSST) [5]. Similarly, within the context of SNNs, Pritchard et al. demonstrate how automated systems can efficiently process dynamic spatio-temporal data through radio frequency interference (RFI) detection. Their study describes the potential of SNNs in time-series functions, achieving competitive performance while discovering meaningful patterns in astronomical datasets [6].

3 Methodology

Our methodology consists of the following stages: data generation, preprocessing, spike encoding, SNN training, anomaly scoring, and evaluation. Primary experiments will use synthetic, astronomical-like time-series signals that simulate normal behavior with injected anomalies such as frequency shifts. Synthetic data provides controlled ground-truth labeling, allowing for the quantitative evaluation of detection accuracy and performance rates. To demonstrate real-world applicability, publicly available datasets of Kepler light curve data will be used for validation in later stages of development [7].

Time-series signals will be normalized to preserve variations relevant to anomaly detection. The processed signals will be encoded into spike trains, using temporal or rate-based encoding standards. This will enable efficient processing by the SNN. The network will be trained on normal data to learn baseline temporal dynamics. Anomalies will be labeled as deviations from expected spike patterns. Anomaly scores will be based on the differences between observed and learned behavior. Performance will be evaluated by using synthetic datasets with known anomalies.

4 References

[1] "Time-Domain Astronomy," Harvard-Smithsonian Center for Astrophysics. `https://www.cfa.harvard.edu/research/topic/time-domain-astronomy`.

[2] V. Etsebeth, M. Lochner, M. Walmsley, and M. Grespan, “Astronomaly at scale: Searching for anomalies amongst 4 million galaxies,” Monthly Notices of the Royal Astronomical Society, vol. 529, no. 1, pp. 732–747, Mar. 2024. doi: 10.1093/mnras/stae496.

[3] K. Yamazaki et al., ”Spiking neural networks and their applications: A review,” Brain Sciences, vol. 12, no. 7, p. 863, Jun. 2022. doi: `https://doi.org/10.3390/brainsci12070863`.

[4] N. J. Pritchard, A. Wicenec, R. Dodson, and M. Bennamoun, “Polarisation-inclusive spiking neural networks for real-time RFI detection in modern radio telescopes,” arXiv preprint arXiv:2504.11720v2, 2025.

[5] D. Muthukrishna, G. Narayan, K. S. Mandel, R. Biswas, and R. Hložek, ”RAPID: Early Classification of Explosive Transients Using Deep Learning,” Publications of the Astronomical Society of the Pacific, vol. 131, no. 1005, p. 118002, Sep. 2019. doi: 10.1088/1538-3873/ab1609. `https://doi.org/10.1088/1538-3873/ab1609`

[6] N. J. Pritchard, A. Wicenec, M. Bennamoun, and R. Dodson, ”Spiking neural networks for radio frequency interference detection in radio astronomy,” Communications Physics, vol. 8, p. 517, 2025. doi: 10.1038/s42005-025-02420-7. `https://doi.org/10.1038/s42005-025-02420-7`

[7] NASA Kepler Mission Archive, Mikulski Archive for Space Telescopes (MAST), Space Telescope Science Institute.